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Problem

Each of m cards is labelled by one of the numbers $1, 2, \dots, m$. Prove that if the sum of the labels of any subset of the cards is not a multiple of $m + 1$, all cards are labelled by the same number.

Solution. Let the cards be labelled with the numbers $c_1, c_2, \dots, c_m$. Let $S_k = c_1 + c_2 + \dots + c_k$.

We know that $m + 1 \mid S_k$ for all $1 \leq k \leq m$. We also know that $S_i \not\equiv S_j \pmod{m + 1}$ for $1 \leq i \neq j \leq m$.

Hence each S_k uniquely represents a remainder modulo $m + 1$.

Consider S_i and $S_{i+1} = S_i + c_{i+1}$. Since our sum with the c_k 's was arbitrary, we can switch c_i and c_{i+1} and we would still have all remainders modulo $m + 1$.

So now we have $S'_i = S_{i-1} + c_{i+1}$ and the rest of the sums remain unchanged for obvious reasons. So, $S'_i \equiv S_i \pmod{m + 1}$. So, $c_{i+1} \equiv c_i \pmod{m + 1}$. Since all of the c_k 's are less than $m + 1$, we must have $c_{i+1} = c_i$. We can conclude this for any i . Hence we have $c_1 = c_2 = \dots = c_m$.